Matrices & Systems

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Given the matrices $A = [4, -2; 1, 3]$ and $B = [-1, 5; 2, 0]$, what is the matrix sum $A + B$?

To add two matrices of the same dimension, add their corresponding entries: - Row 1, Column 1: $4 + (-1) = 3$ - Row 1, Column 2: $-2 + 5 = 3$ - Row 2, Column 1: $1 + 2 = 3$ - Row 2, Column 2: $3 + 0 = 3$ Therefore, $A + B = [3, 3; 3, 3]$. Common misconceptions: - Subtracting entries instead of adding yields $[5, -7; -1, 3]$. - Multiplying corresponding entries entrywise yields $[-4, -10; 2, 0]$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Given the matrices $A = (3, -2; 1, 4)$ and $B = (-5, 2; 0, -3)$, what is the matrix $2A - B$?

To find $2A - B$, first compute the scalar multiple $2A$ by multiplying every entry of A by 2: $2A = 2(3, -2; 1, 4) = (2(3), 2(-2); 2(1), 2(4)) = (6, -4; 2, 8)$ Next, subtract matrix B from $2A$ entry by entry: $2A - B = (6 - (-5), -4 - 2; 2 - 0, 8 - (-3)) = (6 + 5, -6; 2, 8 + 3) = (11, -6; 2, 11)$ Common errors: - Adding the matrices instead of subtracting gives $2A + B = (1, -2; 2, 5)$. - Forgetting to multiply A by the s.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ Substitute the given values into the equation: $6^2 = x(x + 5)$ $36 = x^2 + 5x$ $x^2 + 5x - 36 = 0$ Factor the quadratic equation: $(x + 9)(x - 4) = 0$ This yields $x = -9$ or $x = 4$. Because x represents a geometric length, it must be positive, so $x = 4$. The length of hypotenuse AB is the sum of segments AD and DB.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

Let a_1 be the first term and r be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1 r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1 r + a_1 r^2 + a_1 r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for S_4: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for a_1.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the xy-plane, an angle θ is in standard position. If the terminal side of θ passes through the point $(-5, 12)$, what is the value of $\sin(\theta)$?

To find $\sin(\theta)$ for an angle in standard position whose terminal side passes through $(x, y) = (-5, 12)$: 1. Calculate the distance r from the origin to the point: $r = \sqrt{x^2 + y^2} = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. By the coordinate definition of trigonometric functions: $\sin(\theta) = \frac{y}{r} = \frac{12}{13}$ Misconception checks: - $-\frac{5}{13}$ results from confusing sine with cosine, where $\cos(\theta) = \frac{x}{r}$. - $-\frac{12}{13}$ comes from mistakenly as.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a right-angled triangle, one of the acute angles measures 30°. If the side opposite this angle has a length of 5 units, what is the length of the hypotenuse?

Let the right-angled triangle be denoted as ABC, with the right angle at C. Let angle A be 30°. The side opposite to angle A is BC, and its length is given as 5 units. We are looking for the length of the hypotenuse AB. The sine of an angle in a right-angled triangle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. Therefore, $\sin(A) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{BC}{AB}$. We have $\sin(30^\circ) = \frac{5}{AB}$. We know that.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = (5, -1; 2, 4)$ and $B = (k, 2; -2, 5)$, for what value of the constant k is the matrix $C = 2A - B$ not invertible?

To determine the value of k for which $C = 2A - B$ is not invertible, proceed in three steps: 1. First, calculate the scalar multiple $2A$: $2A = 2(5, -1; 2, 4) = (10, -2; 4, 8)$ 2. Next, subtract matrix B from $2A$ to find C: $C = 2A - B = (10, -2; 4, 8) - (k, 2; -2, 5) = (10 - k, -2 - 2; 4 - (-2), 8 - 5) = (10 - k, -4; 6, 3)$ 3. A square matrix is not invertible if and only if its determinant is equal to zero.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

The general term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n - 1)d$ Substitute $a_1 = 5$, $d = 3$, and $n = 6$: $a_6 = 5 + (6 - 1) \cdot 3 = 5 + 5 \cdot 3 = 5 + 15 = 20$ Misconception Analysis: - 23 comes from using $a_1 + n \cdot d = 5 + 6(3) = 23$, forgetting to subtract 1 from n. - 17 is a_5, resulting from an off-by-one counting error. - 18 comes from directly multiplying $n \cdot d = 6 \cdot 3 = 18$, neglecting the initial term.

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9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What are the coordinates of the center and the radius of the circle given by the equation $(x-4)^2+(y+1)^2=36$?

The standard form of the equation of a circle is $(x-h)^2+(y-k)^2=r^2$, where (h,k) is the center and r is the radius. Matching the given equation $(x-4)^2+(y+1)^2=36$ to the standard form: 1. $x-h=x-4\Rightarrow h=4$ 2. $y-k=y-(-1)\Rightarrow k=-1$ 3. $r^2=36\Rightarrow r=\sqrt{36}=6$ Therefore, the center is $(4,\,-1)$ and the radius is 6. Common misconceptions: - Forgetting the negative signs in $(x-h)$ and $(y-k)$ leads to the incorrect center $(-4,1)$. - Forg.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A small bakery recorded the number of muffins sold each day for a week. The results are shown in the bar chart below. What is the total number of muffins sold during the week?

The bar chart shows the number of muffins sold for each day of the week. To find the total number of muffins sold, we sum the sales for each day: Monday (10) + Tuesday (5) + Wednesday (15) + Thursday (8) + Friday (7) + Saturday (5). Total = 10 + 5 + 15 + 8 + 7 + 5 = 50 muffins.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The polynomial $P(x)=2x^3+ax^2+bx-6$ has a factor of $(x-2)$. When $P(x)$ is divided by $(x+1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x-3)$?

We use the Factor Theorem and the Remainder Theorem: 1. Since $(x-2)$ is a factor of $P(x)$, we have $P(2)=0$: $P(2)=2(2)^3+a(2)^2+b(2)-6=16+4a+2b-6=4a+2b+10=0$ $2a+b=-5\quad---$ (Equation 1) 2. Since the remainder when divided by $(x+1)$ is -12, we have $P(-1)=-12$: $P(-1)=2(-1)^3+a(-1)^2+b(-1)-6=-2+a-b-6=a-b-8=-12$ $a-b=-4\quad---$ (Equation 2) 3. Add Equation 1 and Equation 2 to eliminate b.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

An ellipse in the xy-plane is defined by the equation $9x^2+16y^2-54x+64y+c=0$ where c is a constant. If the length of the minor axis of the ellipse is 6, what is the value of c?

To find c, convert the given equation into standard form by completing the square for both x and y: 1. Group the x and y terms: $(9x^2-54x)+(16y^2+64y)=-c$ 2. Factor out the leading coefficients: $9(x^2-6x)+16(y^2+4y)=-c$ 3. Complete the square inside each set of parentheses: $9(x^2-6x+9)+16(y^2+4y+4)=-c+9(9)+16(4)$ $9(x-3)^2+16(y+2)^2=145-c$ 4. Divide by $145-c$ to write the equation in standard form.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

By the Factor Theorem, if $x = 2$ and $x = -3$ are zeroes of $P(x)$, then $P(2) = 0$ and $P(-3) = 0$. 1. Substitute $x = 2$: $P(2) = (2)^4 + a(2)^3 - 7(2)^2 + b(2) - 12 = 0$

$16 + 8a - 28 + 2b - 12 = 0 \Rightarrow 8a + 2b - 24 = 0 \Rightarrow 4a + b = 12$ (Equation 1) 2. Substitute $x = -3$:

$P(-3) = (-3)^4 + a(-3)^3 - 7(-3)^2 + b(-3) - 12 = 0$ $81 - 27a - 63 - 3b - 12 = 0 \Rightarrow -27a - 3b + 6 = 0 \Rightarrow 9a + b = 2$ (Equation 2) 3. Solve the system of linear equations: Subt.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

The sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$. If the sequence converges to a limit L, then as $n \to \infty$, we have $a_{n+1} \to L$ and $a_n \to L$. Substituting L into the recurrence relation, we get: $L = \frac{L^2 + 4}{2L}$ Multiplying both sides by $2L$ (assuming $Leq0$), we have: $2L^2 = L^2 + 4$ $L^2 = 4$ This gives $L = 2$ or $L = -2$. Let's check the first few terms of the sequence to determine the sign of the limit. $a_1 = 3$..

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A hyperbola is defined by the equation $\frac{(x-h)^2}{16} - \frac{(y-k)^2}{9} = 1$. If the vertices of the hyperbola are located at $(5, 2)$ and $(-1, 2)$, determine the values of h and k.

The standard form of a horizontal hyperbola centered at (h, k) is $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. The vertices of such a hyperbola are located at $(h \pm a, k)$. Given the vertices are $(5, 2)$ and $(-1, 2)$, we can equate these to $(h + a, k)$ and $(h - a, k)$ respectively. From the y-coordinates of the vertices, we see that $k = 2$. From the x-coordinates, we have two equations: 1) $h + a = 5$ 2) $h - a = -1$ Adding these two equations together, we get $(h + a) + (h - a) = 5 + (-1)$, which simplifies to.

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16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

For all real values of x such that $xeq-1$ and $xeq2$, the equation $\frac{ax+7}{x+1}-\frac{b}{x-2}=\frac{2x^2-3x-20}{x^2-x-2}$ is true, where a and b are constants. What is the value of $a+b$?

To find the values of a and b, combine the fractions on the left-hand side over their common denominator, $(x+1)(x-2)=x^2-x-2$: $\frac{(ax+7)(x-2)-b(x+1)}{(x+1)(x-2)}=\frac{2x^2-3x-20}{x^2-x-2}$ Since the denominators are equal for all $xeq-1,2$, their numerators must be identical for all such values of x:
$(ax+7)(x-2)-b(x+1)=2x^2-3x-20$ Expand the left side:
$ax^2-2ax+7x-14-bx-b=2x^2-3x-20$ $ax^2+(7-2a-b)x-(14+b)=2x^2-3x-20$ Equ.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Sequences & Series

A manufacturing process produces items, and the number of defects per batch of 100 items follows a Poisson distribution with a mean of λ. The cost of inspecting a batch is C_i dollars, and the cost of repairing a defective item is C_r dollars. If a batch has k defects, the total cost for that batch is $C(k)=C_i+k\cdot C_r$. If $\lambda=3$, $C_i=50$, and $C_r=10$, find the expected total cost per batch, given that the expected number of defects in a batch is $E[K]=\lambda$.

Let K be the random variable representing the number of defects in a batch of 100 items. K follows a Poisson distribution with mean λ. The cost of inspecting a batch is C_i, and the cost of repairing a defective item is C_r. The total cost for a batch with k defects is $C(k)=C_i+k\cdot C_r$. We are given $\lambda=3$, $C_i=50$, and $C_r=10$. We need to find the expected total cost per batch, $E[C(K)]$. Using the linearity of expectation: $E[C(K)]=E[C_i+K\cdot C_r]$.

18 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Conic Sections

Consider the equation $\sqrt{2x+k}=x-1$, where k is a constant. If $x=5$ is a valid solution to the equation, what is the extraneous solution that arises when solving the equation algebraically by squaring both sides?

Step 1: Find the value of the parameter k by substituting the valid solution $x=5$ into the equation: $\sqrt{2(5)+k}=5-1\Rightarrow\sqrt{10+k}=4$ Squaring both sides gives $10+k=16$, so $k=6$. Step 2: Substitute $k=6$ back into the equation and solve by squaring both sides: $\sqrt{2x+6}=x-1$
$2x+6=(x-1)^2=x^2-2x+1$ $x^2-4x-5=0$ Factoring the quadratic equation gives:
$(x-5)(x+1)=0\Rightarrow x=5$ or $x=-1$ Step 3: Check for extraneous sol.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Trigonometric Functions

A set of exam scores is normally distributed with a mean of $\mu = 74$ and a standard deviation of $\sigma = 8$. What is the z-score corresponding to an exam score of 58?

To find the z-score corresponding to a given data value, use the standard normal distribution formula: $z = \frac{x - \mu}{\sigma}$ where x is the individual score, μ is the population mean, and σ is the standard deviation. Substitute the given values into the formula: $z = \frac{58 - 74}{8} = \frac{-16}{8} = -2$ A z-score of -2 indicates that the score of 58 is located 2 standard deviations below the mean. Common Misconceptions: - 2: Results from subtracting the score from the mean.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs

Let $f(x) = \sqrt{x^2 - 6x + 9}$ and $g(x) = x - 3$. For which values of x is $f(x) = g(x)$?

First, simplify the expression for $f(x)$. We recognize that the radicand, $x^2 - 6x + 9$, is a perfect square trinomial, which can be factored as $(x - 3)^2$. Therefore, $f(x) = \sqrt{(x - 3)^2}$. The definition of the square root of a square is $\sqrt{a^2} = |a|$. Applying this, we get $f(x) = |x - 3|$. Now, we need to solve the equation $f(x) = g(x)$, which becomes $|x - 3| = x - 3$. The absolute value of an expression is equal to the expression itself if and only if the expression is non-negative. Thus, $|x - 3| = x - 3$ if.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in reciprocal, quotient, and pythaogrean identities, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 6$ in cofunction identities, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angle sum and difference identities, $m\angle A = 47^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 88° ✓

b. 98°

c. 80°

d. 133°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (47^\circ + 45^\circ) = 88^\circ$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for reciprocal, quotient, and pythaogrean identities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for cofunction identities consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 14

c. 12 ✓

d. 17

Substituting $x = 1$ into $y = 5x + 7$ gives $y = 5(1) + 7 = 12$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for verifying trig identities consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 8$ in verifying trig identities, evaluate $f(3)$.

a. 23 ✓

b. 28

c. 15

d. 27

Substitute $x = 3$: $f(3) = 5(3) + 8 = 23$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For eigenvalue λ with eigenvector v, any scalar multiple cv is also an eigenvector. True or False?

a. Only if $c = 1$

b. False

c. Only if $c > 0$

d. **True** ✓

$A(cv) = c(Av) = c\lambda v = \lambda(cv)$. ✓.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the matrix equation $(2,1,-2;4,-1,-1;2,-2,1)(x;y;z)=(9;9;k)$ where k is a real constant. Given that the system has infinitely many real solutions (x,y,z), what is the minimum possible value of $x^2+y^2+z^2$?

a. **9** ✓

b. 3

c. 6

d. 45

We solve the problem in three steps: determine k for consistency, parameterize the infinite solution set, and minimize the quadratic objective function. **Step 1: Determine k for consistency** Write the augmented matrix of the system: $(2,1,-2,9;4,-1,-1,9;2,-2,1,k)$ Perform Gaussian elimination: - Replace Row 2 with R_2-2R_1: $(4-4,\ -1-2,\ -1-(-4)\mid 9-18)=(0,-3,3\mid -9)$, which simplifies to $(0,1,-1\mid 3)$. - Replace Row 3 with R_3-R_1.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the system of linear equations represented by the matrix equation $(1,1,1;1,2,3;3,5,k)(x;y;z)=(6;14;m)$ where k and m are real constants. Suppose the system is consistent and has infinitely many solutions. Let $P(x,y,z)$ be the unique solution that minimizes the squared Euclidean distance from the origin, $d^2=x^2+y^2+z^2$. What is the value of $k+m+d^2$?

a. **55** ✓

b. 47

c. 38

d. 109

We solve the problem in three main stages: 1. **Determine the parameter k:** For the system to have infinitely many solutions, the coefficient matrix must be singular (its determinant must be zero): $\det(1,1,1;1,2,3;3,5,k)=1(2k-15)-1(k-9)+1(5-6)=k-7$ Setting $k-7=0$ yields $k=7$. 2. **Determine the parameter m:** Set up the augmented matrix with $k=7$ and perform Gaussian elimination.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $A = (x, 2; 3, x - 1)$. If A is not invertible and $x > 0$, what is the value of $\det(A + 3I)$, where I is the 2×2 identity matrix?

a. **24** ✓

b. 9

c. 0

d. -6

A square matrix is not invertible if and only if its determinant is zero. First, compute $\det(A)$: $\det(A) = x(x - 1) - (2)(3) = x^2 - x - 6$ Setting $\det(A) = 0$: $x^2 - x - 6 = (x - 3)(x + 2) = 0$ Since we are given that $x > 0$, we must have $x = 3$. Next, substitute $x = 3$ into A: $A = (3, 2; 3, 2)$ Now find the matrix $A + 3I$: $A + 3I = (3, 2; 3, 2) + 3(1, 0; 0, 1) = (6, 2; 3, 5)$ Finally, calculate $\det(A + 3I)$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the matrix equation: $(3, -1; 2, k)(x; y) = (5; 7)$. If the matrix $(3, -1; 2, k)$ is singular, what is the value of k?

a. $-3/2$ ✓

b. $3/2$

c. $-2/3$

d. $2/3$

A matrix is singular if and only if its determinant is zero. The determinant of a 2x2 matrix $(a, b; c, d)$ is given by $ad - bc$. For the given matrix $(3, -1; 2, k)$, the determinant is $(3)(k) - (-1)(2) = 3k + 2$. For the matrix to be singular, the determinant must be zero: $3k + 2 = 0$. Solving for k: $3k = -2$, which gives $k = -2/3$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given the matrix $A = (2, 1; 3, 4)$ and the matrix $B = (1, -1; 0, 2)$, what is the resulting matrix $A + B$?

a. $(3, 0; 3, 6)$ ✓

b. $(2, -1; 3, 2)$

c. $(1, 2; 3, 2)$

d. $(3, 1; 3, 4)$

To find the sum of two matrices, we add the corresponding elements. Given matrix A: $A = (2, 1; 3, 4)$ Given matrix B: $B = (1, -1; 0, 2)$ Matrix A + Matrix B = $(2 + 1, 1 + (-1); 3 + 0, 4 + 2)$ Matrix A + Matrix B = $(3, 0; 3, 6)$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given the matrix $A = (2, 1; 3, 4)$ and the matrix $B = (1, 0; 0, 1)$, what is the resulting matrix when A is added to B?

a. $(2, 1; 3, 4)$

b. $(3, 1; 3, 5)$ ✓

c. $(1, 1; 3, 3)$

d. $(2, 0; 0, 4)$

To add two matrices, we add their corresponding elements. \\ Given $A = (2, 1; 3, 4)$ and $B = (1, 0; 0, 1)$, the sum $A + B$ is calculated as follows:\\ $A + B = (2 + 1, 1 + 0; 3 + 0, 4 + 1) = (3, 1; 3, 5)$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove that the determinant of a matrix with two identical rows is zero.

a. Only for 2×2

b. Only if symmetric

c. **Swapping the two identical rows changes the sign of the det but doesn't change the matrix:** $\det = -\det \Rightarrow \det = 0$**.** ✓✓

d. Not true

Applying Matrices & Systems principles, Row swap property. Therefore, the verified correct option is Swapping the two identical rows changes the sign of the det but doesn't change the matrix: $\det = -\det \Rightarrow \det = 0$. ✓.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Conic Sections

State the rank-factorization theorem.

a. $A = A^T A$

b. Only for square matrices

c. Not always possible

d. **Any $m \times n$ matrix A of rank r can be written as $A = CR$ where C is $m \times r$ and R is $r \times n$** ✓

Applying Matrices & Systems principles, Rank factorization theorem. Therefore, the verified correct option is Any $m \times n$ matrix A of rank r can be written as $A = CR$ where C is $m \times r$ and R is $r \times n$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Trigonometric Functions

State the Sylvester rank inequality: for $A_{m \times n}$ and $B_{n \times p}$:

a. $\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$

b. $\text{rank}(AB) = \text{rank}(A) + \text{rank}(B)$

c. $\text{rank}(A) + \text{rank}(B) - n \le \text{rank}(AB) \le \min(\text{rank}(A), \text{rank}(B))$ ✓

d. $\text{rank}(AB) = 0$

Applying Matrices & Systems principles, Sylvester inequality. Therefore, the verified correct option is $\text{rank}(A) + \text{rank}(B) - n \le \text{rank}(AB) \le \min(\text{rank}(A), \text{rank}(B))$.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Functions & Their Graphs

Prove that $\text{rank}(A) + \text{nullity}(A) = n$ (rank-nullity theorem).

a. **By the fundamental theorem of linear algebra: column space dim + null space dim = number of columns ✓**

b. Not true

c. Only for 2×2

d. Only for square matrices

Applying Matrices & Systems principles, Rank-nullity theorem. Therefore, the verified correct option is By the fundamental theorem of linear algebra: column space dim + null space dim = number of columns.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability & Statistics

Prove: for invertible A, the eigenvalues of A^{-1} are $1/\lambda_i$.

a. **If $Av = \lambda v$ then $v = \lambda A^{-1}v \Rightarrow A^{-1}v = (1/\lambda)v$. ✓✓**

b. Only for diagonal matrices

c. Only if $\lambda > 0$

d. Not true

Applying Matrices & Systems principles, Direct proof from definition. Therefore, the verified correct option is If $Av = \lambda v$ then $v = \lambda A^{-1}v \Rightarrow A^{-1}v = (1/\lambda)v$. ✓.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Show that if A has eigenvalue λ, then A^n has eigenvalue λ^n.

a. Not always true

b. Only for symmetric A

c. Use Cayley-Hamilton

d. $A^n v = A^{n-1}(Av) = A^{n-1}(\lambda v) = \lambda A^{n-1} v = \cdots = \lambda^n v$. ✓ ✓

Applying Matrices & Systems principles, Inductive argument. Therefore, the verified correct option is $A^n v = A^{n-1}(Av) = A^{n-1}(\lambda v) = \lambda A^{n-1} v = \cdots = \lambda^n v$. ✓.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Polynomial Functions

State the Cayley-Hamilton theorem.

a. Only 2×2 matrices

b. Only diagonalizable matrices satisfy it

c. **Every square matrix satisfies its own characteristic equation** ✓

d. It applies to vectors

Applying Matrices & Systems principles, Cayley-Hamilton theorem. Therefore, the verified correct option is Every square matrix satisfies its own characteristic equation.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rational Expressions

Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for any $m \times n$ matrices A and $n \times m$ matrix B.

a. Only if $A = B$

b. Not true in general

c. $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji} = \operatorname{tr}(BA)$. ✓ ✓

d. Only for square matrices

Applying Matrices & Systems principles, Cyclic property of trace. Therefore, the verified correct option is $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji} = \operatorname{tr}(BA)$. ✓.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the recurrence $(x_{n+1}; y_{n+1}) = A(x_n; y_n)$, the solution is:

a. $(x_n; y_n) = nA(x_0; y_0)$

b. $(x_n; y_n) = e^{An}(x_0; y_0)$

c. Need eigenvalues first

d. $(x_n; y_n) = A^n(x_0; y_0)$ ✓

Applying Matrices & Systems principles, Matrix recurrence solution. Therefore, the verified correct option is $(x_n; y_n) = A^n(x_0; y_0)$.

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